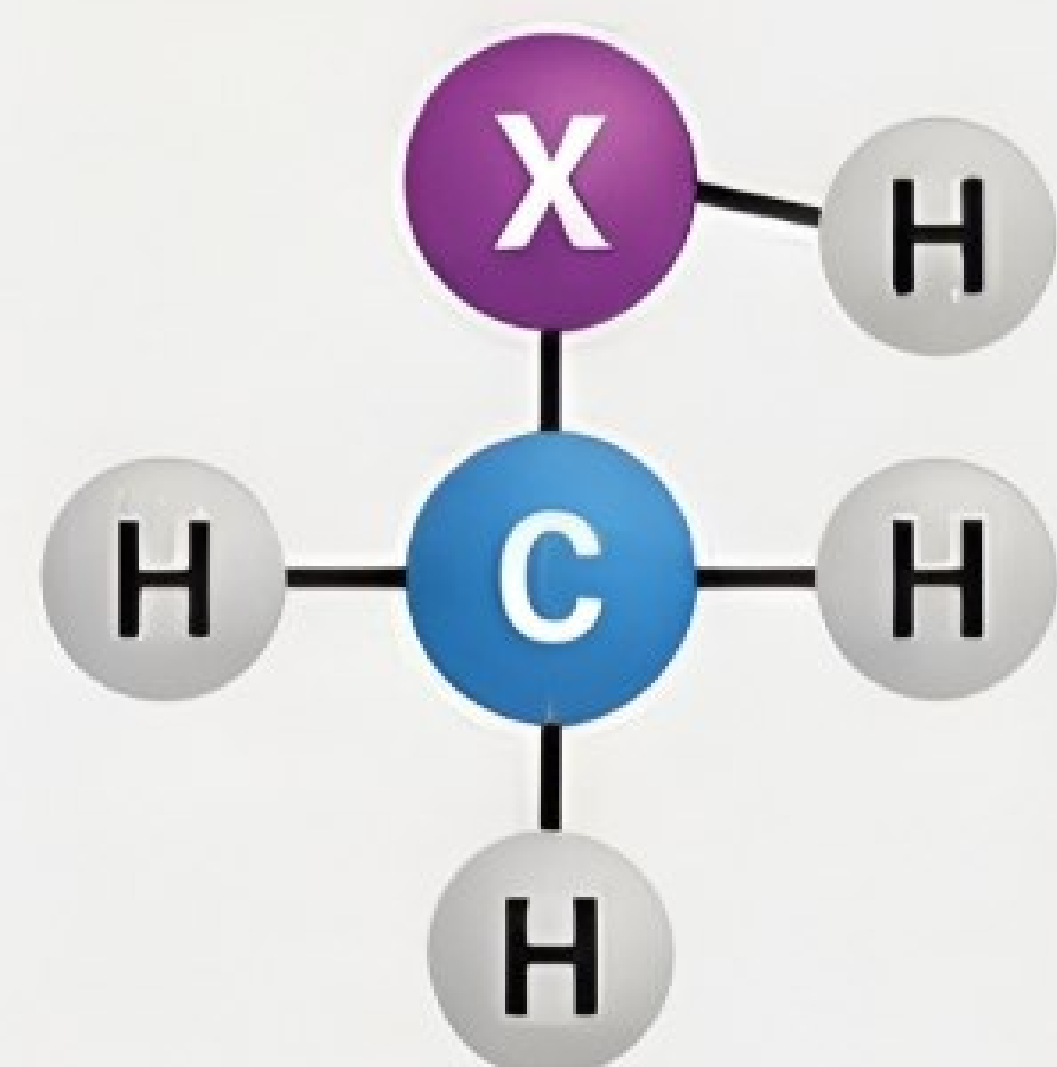


A Visual Guide to Haloalkanes & Haloarenes

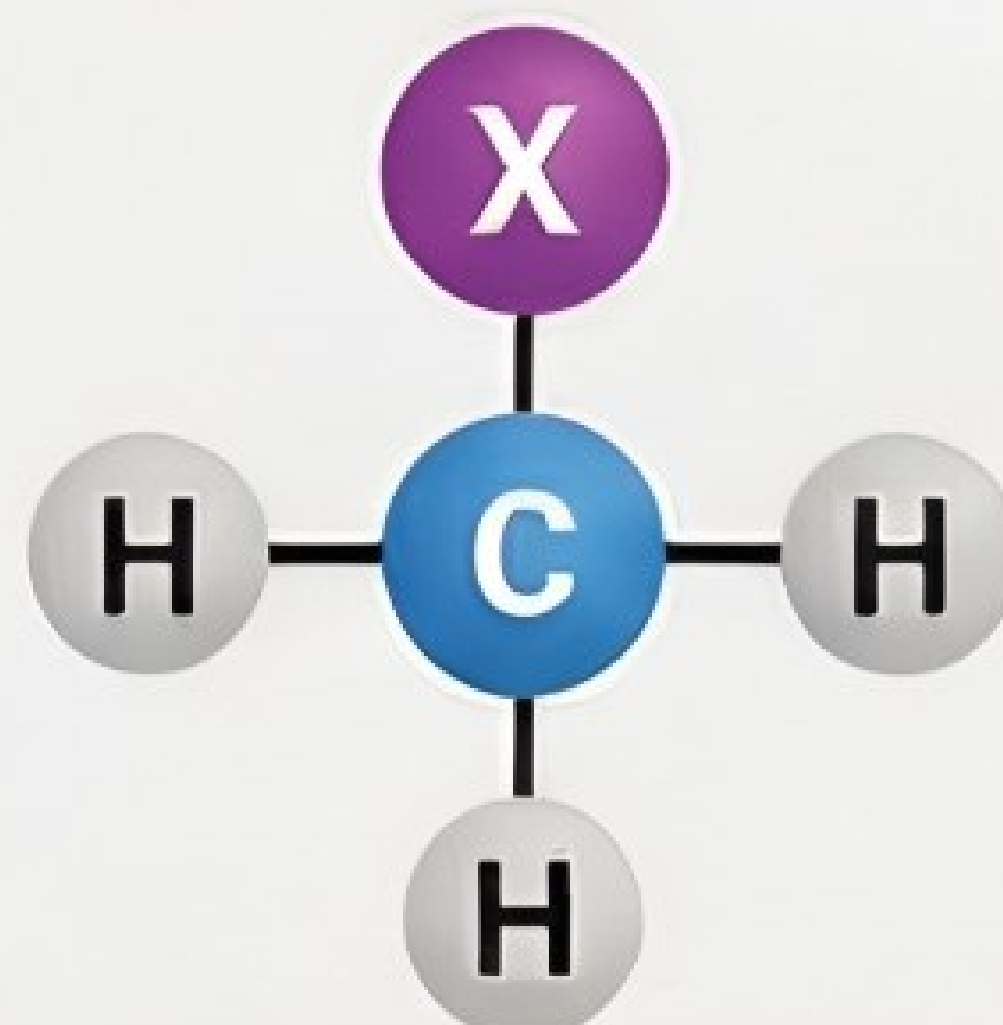
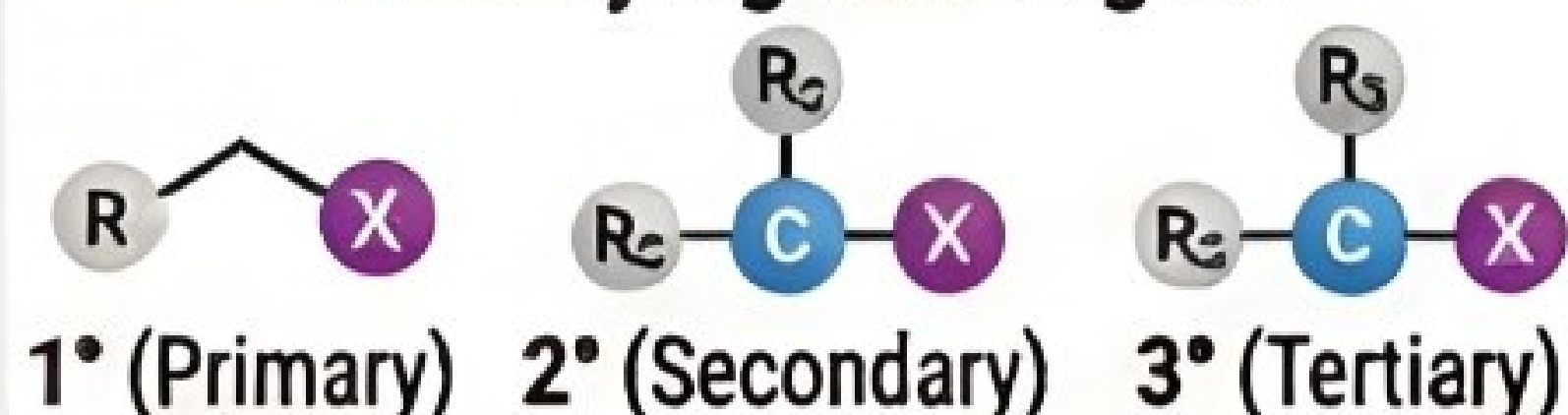
The Basics: Identifying the Compounds



Haloalkane

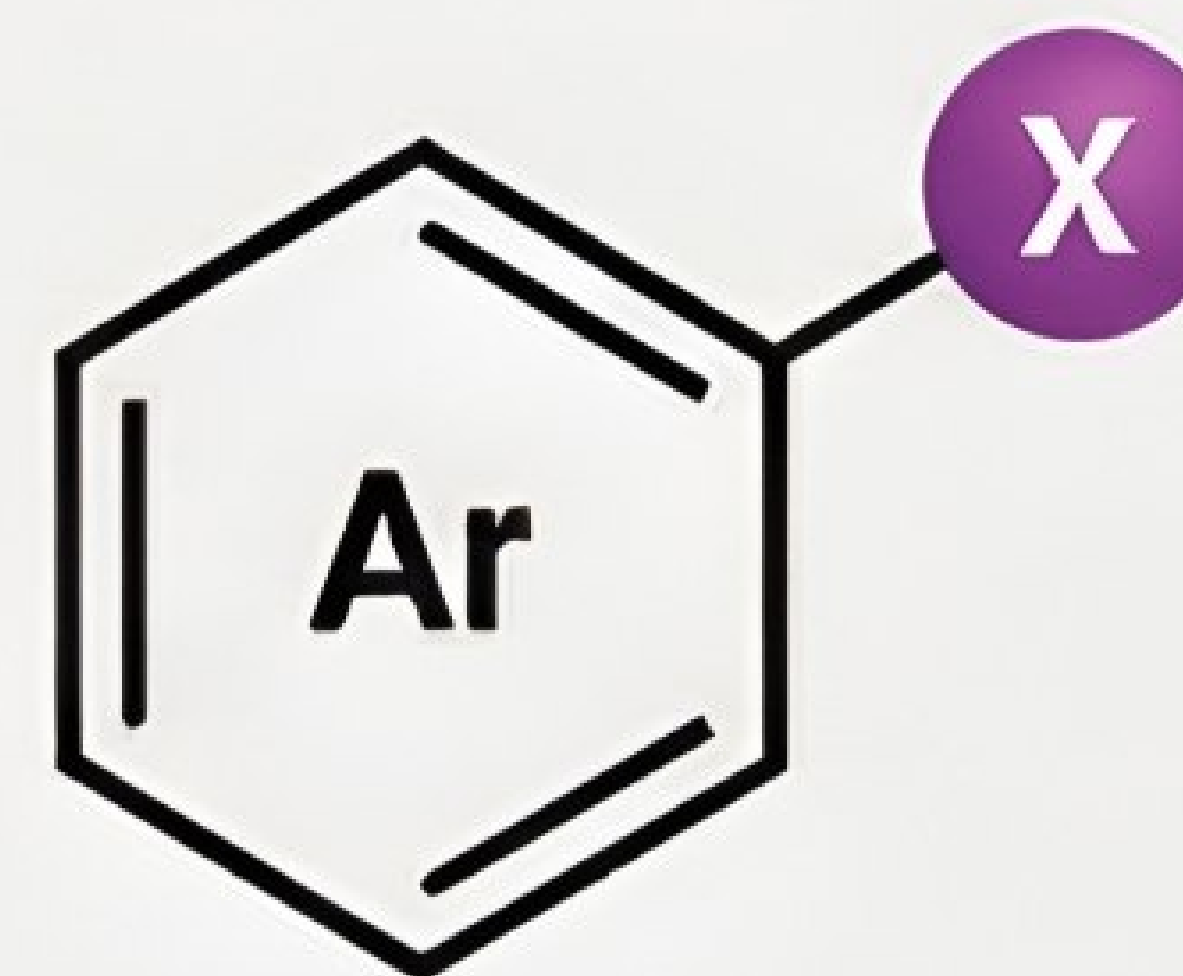
Alkane with one or more hydrogens replaced by a halogen (F, Cl, Br, I).
General formula: R-X.

Identifying the Degree



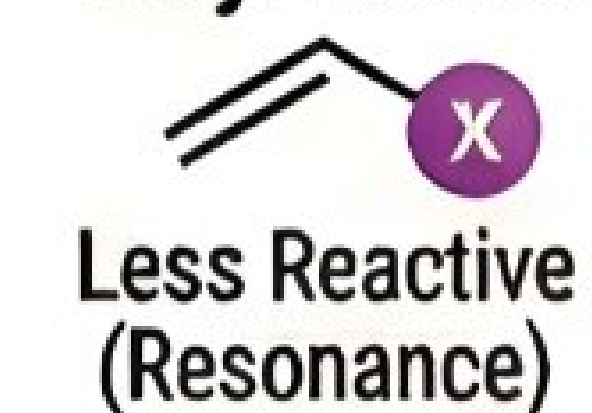
Haloarene

Aromatic compound (like benzene) where a halogen is directly attached.
General formula: Ar-X.

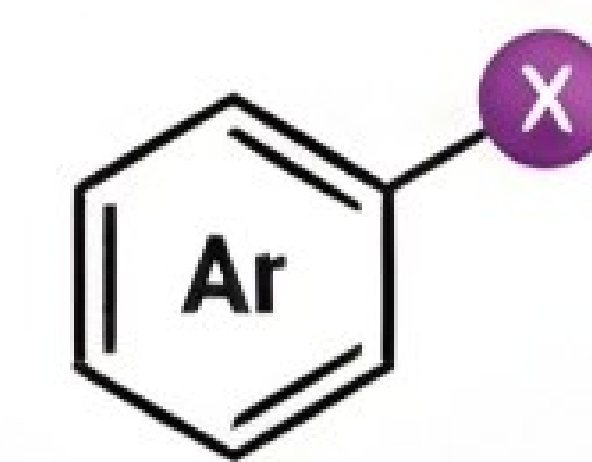
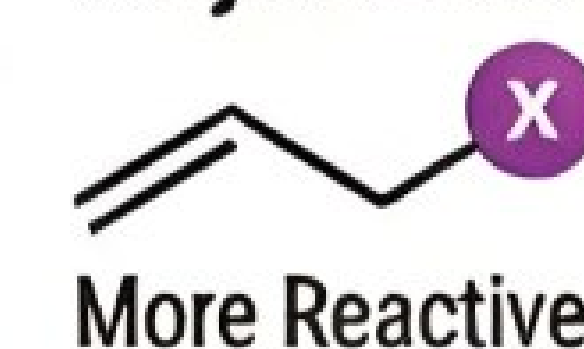


Know Your Halides: Vinyl vs. Allyl

Vinyl Halide



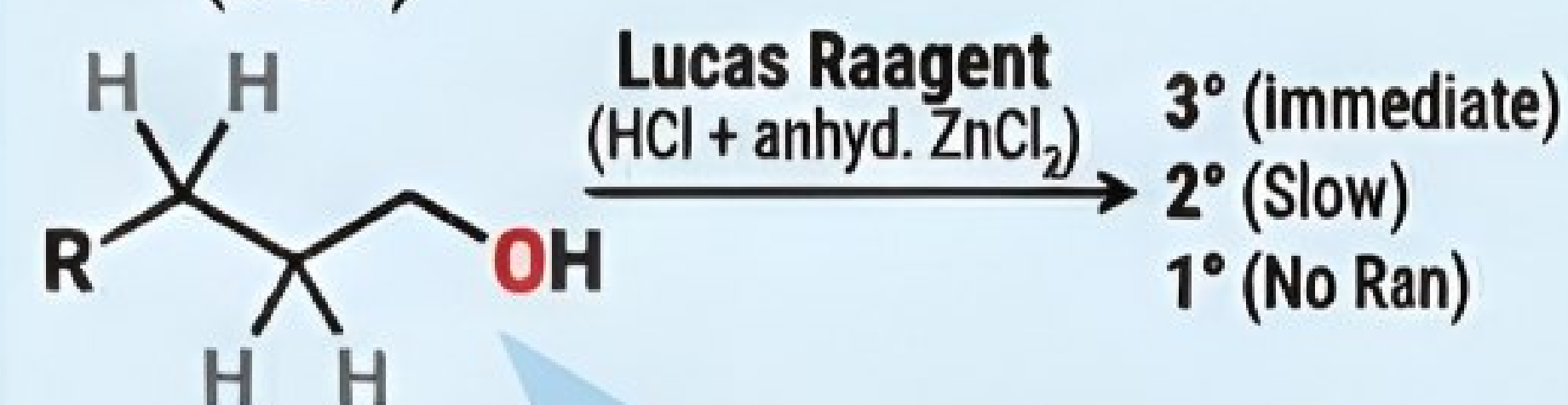
Allyl Halide



Benzyl Halide
Highly Reactive

How to Make Haloalkanes (Preparation Methods)

From Alcohols (R-OH)



The Best Method:
Thionyl Chloride (SOCl₂)
→ Preferred
(SO₂ & HCl are gases = Pure product)

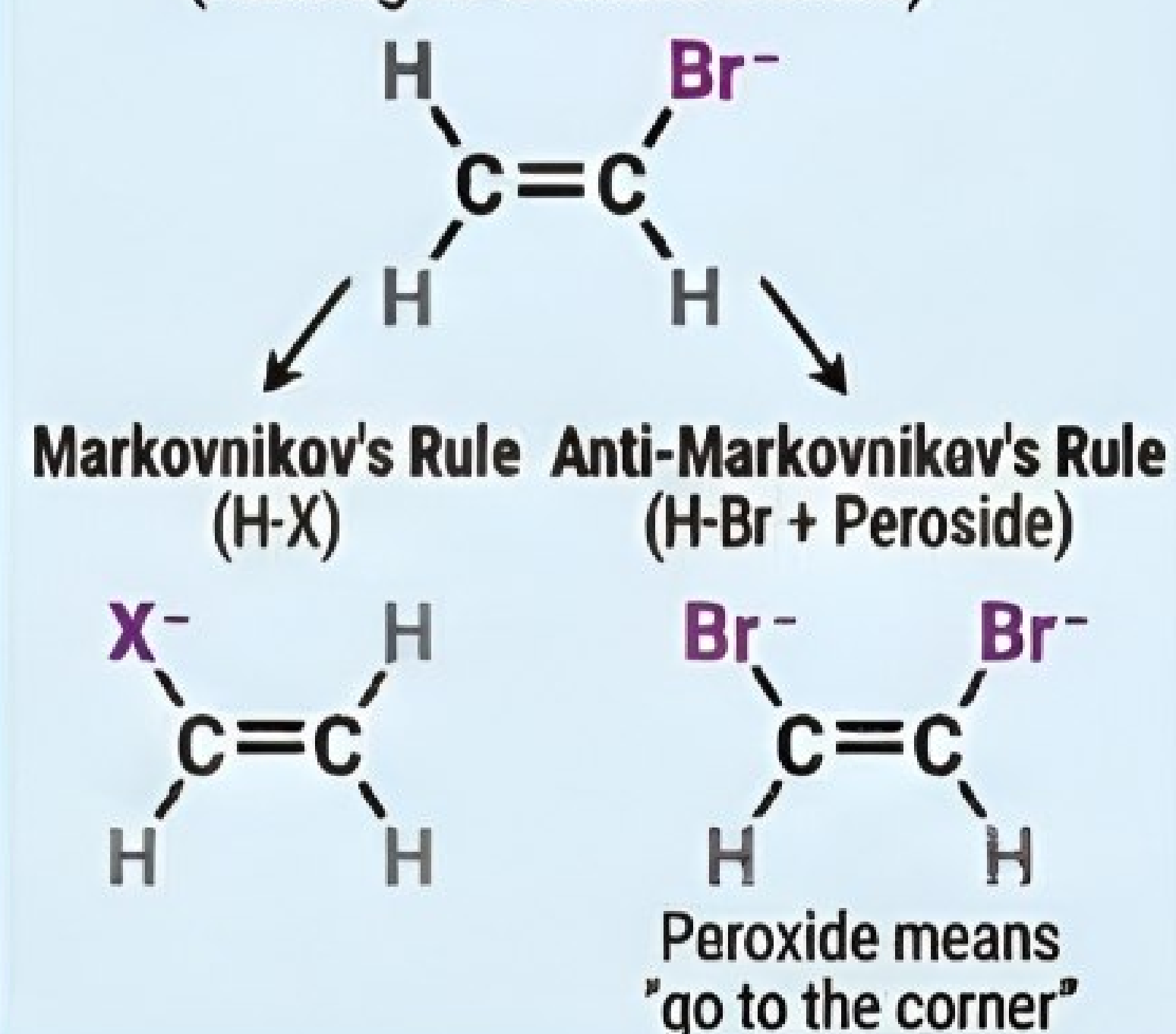
PCl₂

PCl₂

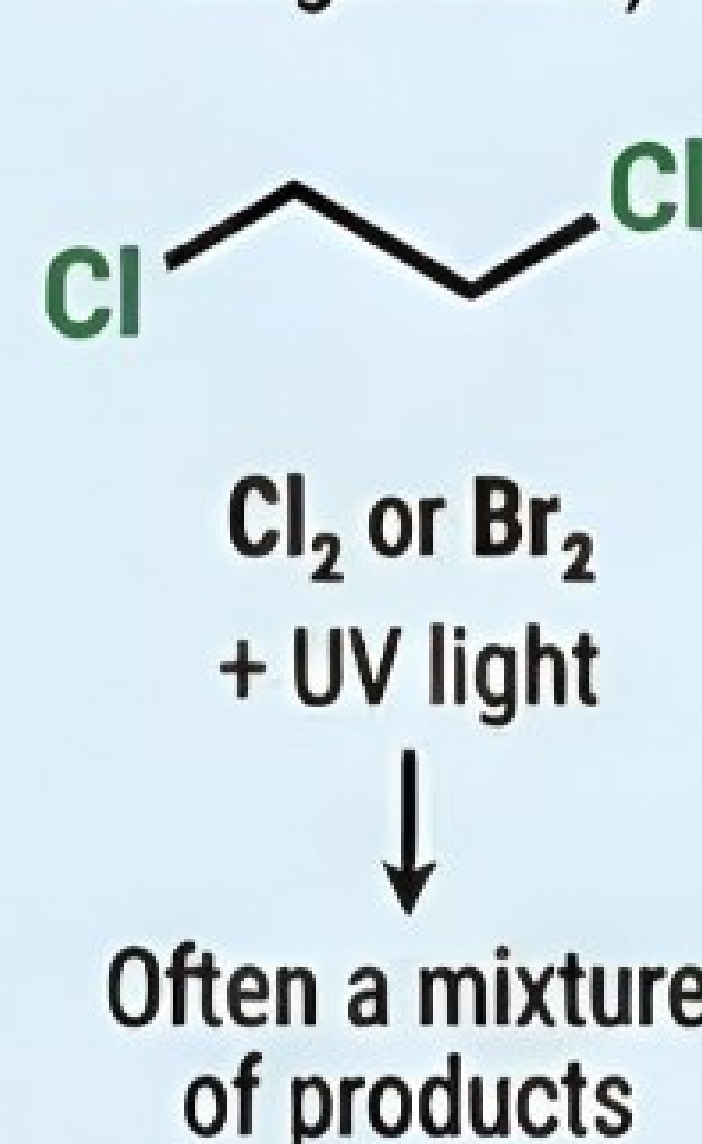
Red P/X₂

NaBr + H₂SO₄

From Alkenes (Adding to Double Bonds)



From Alkanes (Free-Radical Halogenation)



Key Mechanisms & Named Reactions

S_N1 vs. S_N2 Reaction Mechanisms

Steps: One Step

Intermediate: Carbocation formed —C⁺

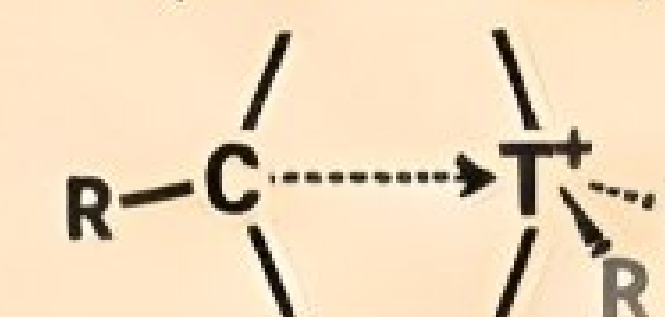
Reactivity Order: 3° > 2° > 1° (Stable C⁺)

Deciding Factor: Steric Hindrance (Crowding)

Solvent: Polar Protic (e.g., H₂O)

S_N1

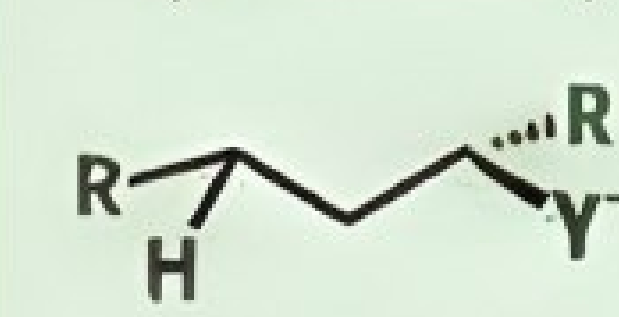
(Unimolecular)



Transition state (No intermediate)

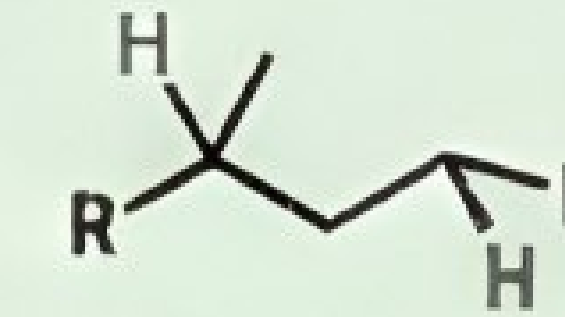
S_N2

(Bimolecular)



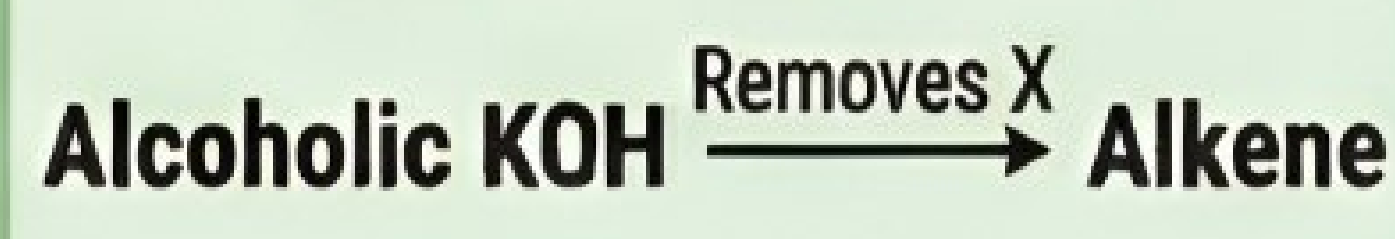
Polar Aprotic (e.g., DMF)

Stereochemistry:



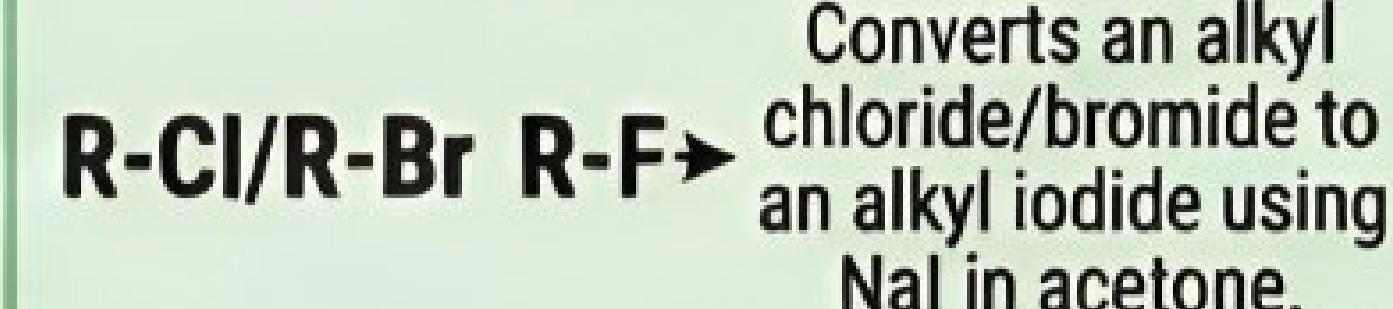
Complete Inversion

Elimination Reaction (Dehydrohalogenation)



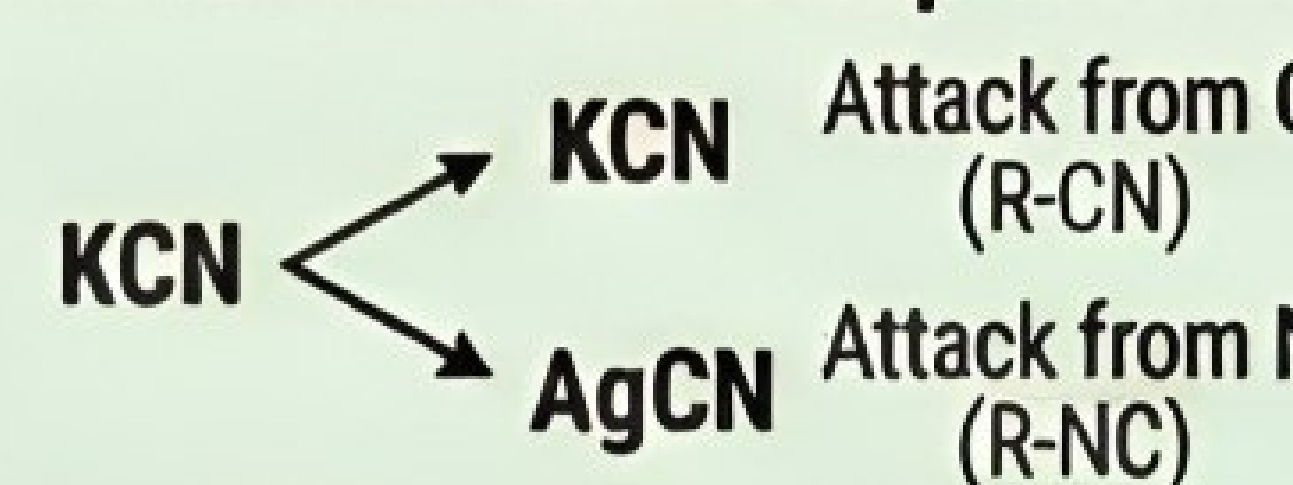
(Mnemonic: Finkelstein uses NaI in Acetone)

Finkelstein Reaction (Halogen Exchange)



(Mnemonic: Swarts = Swag = AgF (Ag Fire))

The Ambident Nucleophile Rule



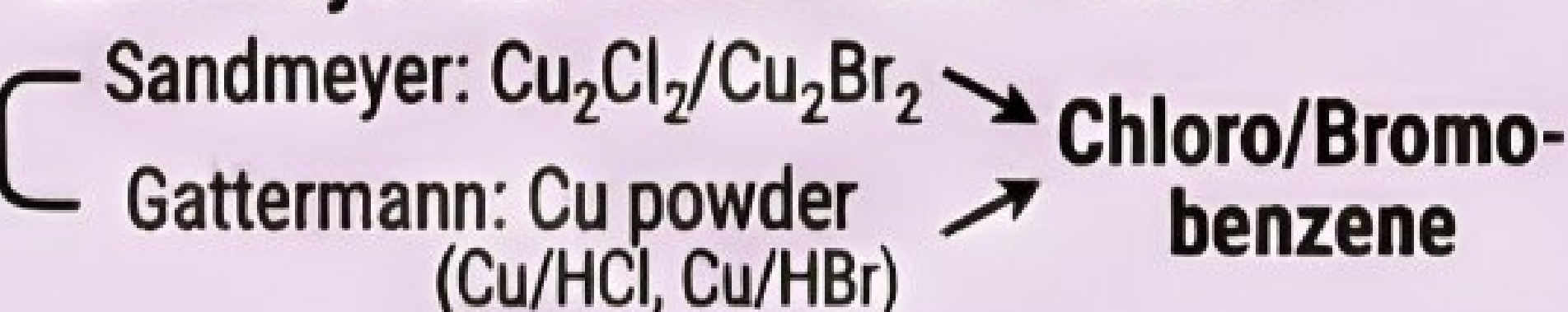
(Mnemonic: When Ag is present, Nitrogen attacks)

Haloarenes: Preparation & Reactivity

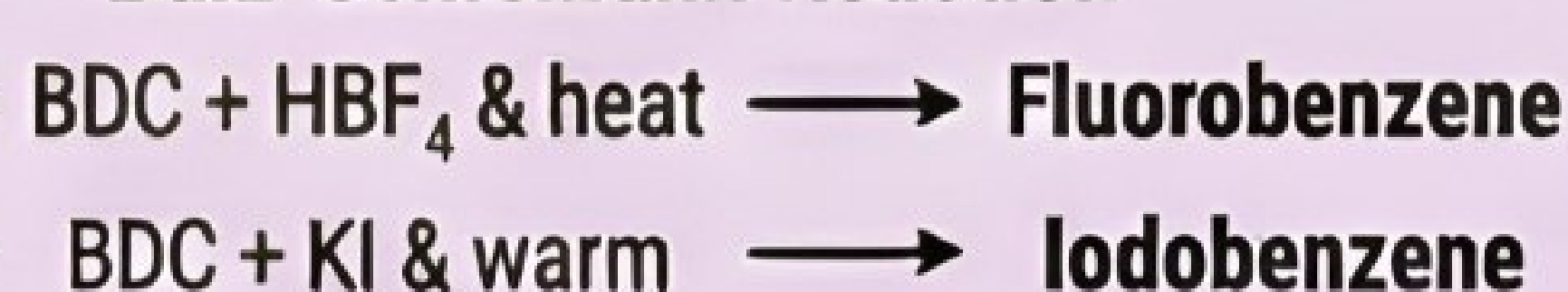
Making Haloarenes from BDC ("The Hero")



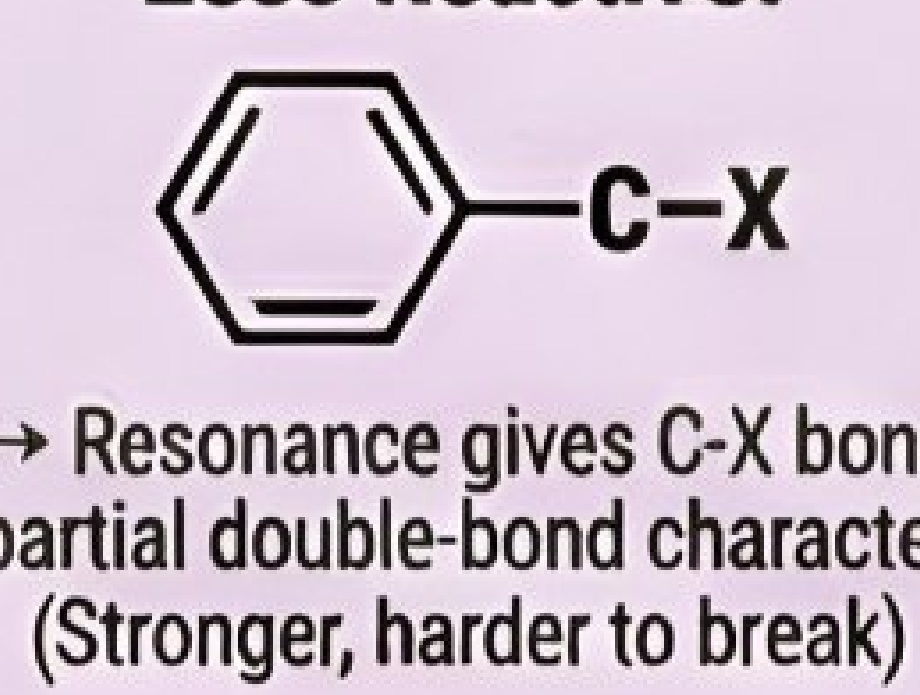
Sandmeyer vs. Gattermann Reaction



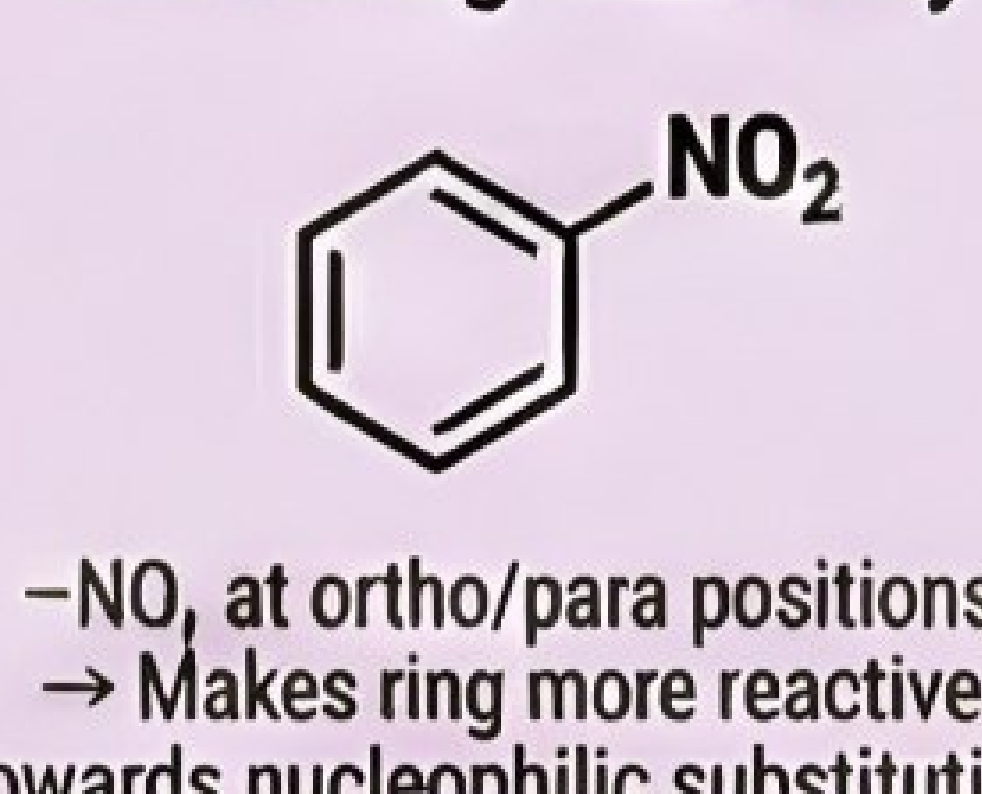
Balz-Schiemann Reaction



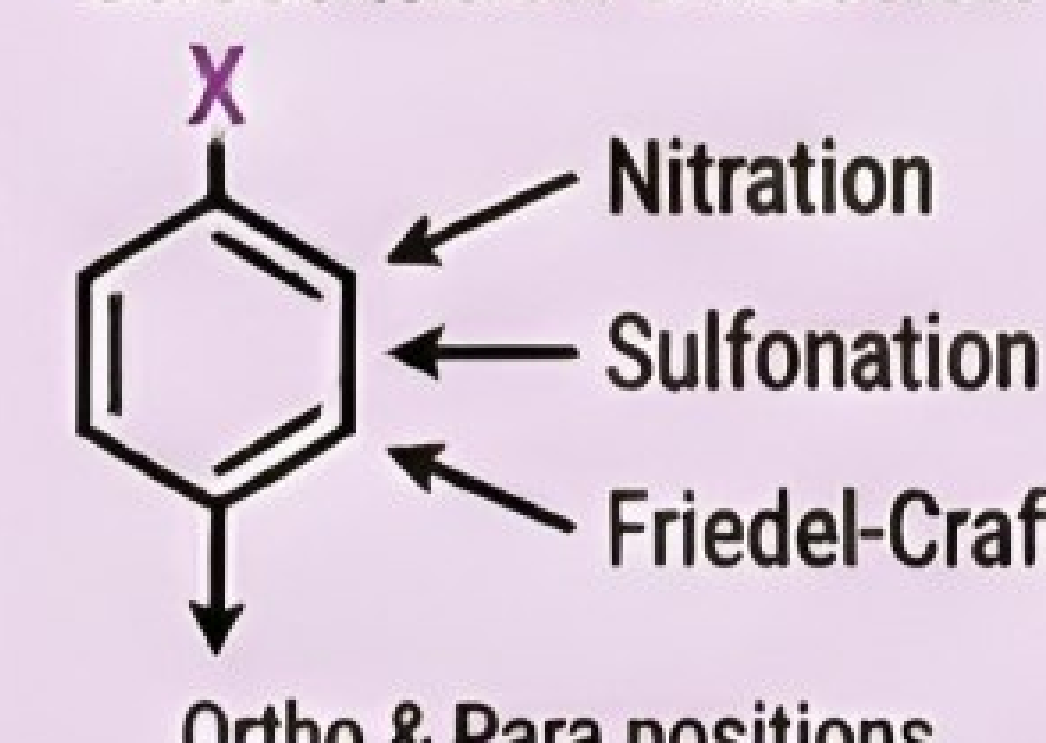
Why are Haloarenes Less Reactive?



Increasing Reactivity



Electrophilic Substitution Direction



Reactions with Sodium (Na)

